# HAMMAD ALI

## Full Stack Software Engineer

**React · Next.js · TypeScript · NestJS · .NET Core · PostgreSQL**

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## PROFESSIONAL SUMMARY

Full Stack Software Engineer specializing in **React**, **Next.js**, **TypeScript** and **NestJS**, with 3+ years delivering enterprise ERP, SaaS and supply-chain platforms across the full development lifecycle. Owned end-to-end delivery across **7+ production systems**, migrated **3 legacy VB.NET systems** to a modern stack, and built a document extraction pipeline on the **OpenAI API** that cut processing time by **70%**. Experienced in requirements analysis, testing, code review, production support and mentoring junior developers.

## EXPERIENCE

### Software Engineer

May 2023 - Present

**Interactive Technologies Gateway**

- Own end-to-end delivery across **7+ enterprise systems** for supply-chain and quality-inspection clients, from React/Next.js frontends to Node.js/NestJS APIs.
- Design backend APIs in **NestJS** and **Prisma ORM** covering authentication, **RBAC**, and multi-module business logic for ERP platforms.
- Migrated **3 legacy VB.NET systems** to React + Node.js + PostgreSQL, cutting **UI load times by 40%** and retiring an unmaintainable codebase.
- Built **Excel Genie**, a reporting engine reused across **10+ ERP modules** in place of per-module reporting code.
- Engineered **AG Grid** interfaces over **50,000+ row datasets** with real-time **WebSocket** synchronization and zero UI lag.
- Created reusable **React component libraries** and NestJS service patterns, cutting **per-module development time by 30%**.
- Ship production endpoints and **Docker**-based services independently through **CI/CD** pipelines.
- Write **unit and integration tests** with **Vitest**, and run **code reviews** across the team before work is merged.
- Own **requirements analysis, estimation and planning** for assigned modules, working directly with business stakeholders.
- Mentor **3-5 junior developers** in application architecture, debugging and code quality.
- Investigate and resolve **production issues** across frontend, API, database and deployment layers, from logs and traces to root cause.

**Cortex** AI-Powered ERP System | React.js, NestJS, PostgreSQL, WebSockets, OpenAI, AG Grid

- Led full-stack delivery across **8 interconnected modules** serving **100+ users across 7+ permission roles**, with admin panels, permission flows and real-time 50,000+ row data grids.
- Built a bulk **PDF extraction pipeline on the OpenAI API** with a per-client template covering **8-12 document formats**, routed through human review before commit: **~70% less processing time**.
- Moved the pipeline off **Azure Document Intelligence** after custom-model training per document type failed to reach usable accuracy.

**Integra** Quality Inspection Platform | React.js, Node.js, PostgreSQL, WebSockets, PDF Generation

- Built **7 end-to-end inspection workflows** with approvals modelled as an explicit **state machine**, real-time status updates, server-side PDF reports and live KPI dashboards.
- Modelled the full reporting hierarchy for **50+ inspectors** through their managers and department head, including finance sign-off on **conveyance claims**.

**Vendor Portal** Supplier Invoicing | React, .NET Core Web API, SQL Server, PDF Automation

- Replaced emailed supplier PDFs with direct uploads, automated extraction and a validation step before commit: **10,000+ documents a year**, **70% less manual effort**.

### Full Stack Developer (Remote)

Jun 2025 - Feb 2026

**Consulting Service International Ltd**

- Built client websites, portals and a live subscription platform in **React**, **Next.js** and **Express**, working remotely from requirements through to deployment.

**Law in Action** Global Regulations Platform | React, Express, Prisma, PostgreSQL lawintoaction.com

- Built a live subscription compliance platform with regulation tracking, **HS-code classification** over the **10,300-entry UN Comtrade set** held in memory, and a metered external API clients pull into their own ERPs.
- Put translation behind a **provider abstraction** over DeepL, Azure, Google and MyMemory with automatic fallback, so one provider's outage never stalls compliance work.

## PERSONAL PROJECTS

**Okulium** Multi-Tenant School Management SaaS | React, Vite, NestJS, Prisma, PostgreSQL, PWA okulium-web.onrender.com

- Shipped a live multi-tenant platform for school groups: attendance, exams and results, fees, timetabling, staff and reporting, as an **installable PWA**.
- Attendance edits write to an **edit-history table** and result disputes are a separate model, so corrections stay auditable.

## TECHNICAL SKILLS

**Frontend:** React.js, Next.js, React Native, TypeScript, JavaScript (ES6+), Redux, Zustand, Tailwind CSS, AG Grid
**Backend & APIs:** Node.js, NestJS, Express.js, .NET Core Web API, Prisma ORM, REST APIs, WebSockets, JWT, RBAC
**Databases & Caching:** PostgreSQL, Microsoft SQL Server, MongoDB, Redis
**Testing & DevOps:** Vitest, unit & integration testing, Docker, CI/CD, Git, GitHub, Vercel
**Engineering Practices:** Requirements analysis, estimation, code reviews, debugging, production support, mentoring
**AI & Automation:** OpenAI API, Gemini, bulk PDF extraction, document intelligence, automation pipelines

## EDUCATION & CERTIFICATIONS

**BS Software Engineering** - Virtual University of Pakistan — Oct 2021 - Oct 2025
**MERN Stack & Frontend Development** - Jawan Pakistan — Jun 2022 - Jun 2023